

# YONGHAO TAN

Ph.D. Candidate in Electronic and Computer Engineering, The Hong Kong University of Science and Technology (HKUST)  
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## EDUCATION

### Ph.D. in Electronic and Computer Engineering

Sept. 2023 - Present

### The Hong Kong University of Science and Technology (HKUST)

Hong Kong, China

Supervisor: Prof. Tim Kwang-Ting Cheng

### B.E. in Microelectronics

Sept. 2019 - Jun. 2023

### Southern University of Science and Technology (SUSTech)

Shenzhen, Guangdong, China

Supervisor: Prof. Fengwei An

## RESEARCH EXPERIENCE

### 5nm UC1e-Enabled Multi-Chiplet Generalizable Rendering Processor

Mar. 2024 - Sept. 2025

AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China

- Designed the architecture and execution scheduler for a 5nm four-chiplet GeNeRF processor, and built an early-stage simulation framework to evaluate buffer management, cross-die source-view caching, dense/sparse D2D transfer, multi-level sparsity control, and hybrid GeNeRF-SR dataflow.
- Developed algorithm-hardware co-design simulation models for projection-area-driven source-view placement, dynamic patch grouping, dense/sparse D2D modes, source-view pruning, tile-level fine-stage sharing, and patch-grained SR routing; used them to validate accuracy-sensitive sparsity and SR decisions and guide hardware-aware end-to-end quantization for silicon deployment.
- Contributed to the implementation of inter-chiplet transfer, chip-level control, and nonlinear computation modules; participated in chip testing and system validation of the fabricated 5nm MCM processor, which integrates the four-chiplet GeNeRF engine in a 45 mm x 45 mm package and achieves 91.43 TOPS/W and 55.43 FPS throughput.

### 55nm ReRAM-on-Logic Stacked LLM Accelerator

Apr. 2024 - Aug. 2025

AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China

- Contributed to algorithm-hardware co-design optimizations for a 55nm ReRAM-on-logic stacked edge LLM accelerator, targeting decoding-stage memory and scheduling bottlenecks with local-rotation-based W4A8 quantization, ReRAM-resident block-clustered codebook reconstruction, and adaptive speculative decoding.
- Implemented and validated the end-to-end algorithm flow, including layer-wise quantized LLM evaluation, codebook-based draft-model weight reconstruction, acceptance/rejection-aware speculative decoding analysis, and hardware-mapping studies that balance target-model EMA reduction against rejected-draft overhead.
- Designed RTL for nonlinear computation modules and related control/datapath logic; assisted chip testing and validation of the 55nm logic die stacked with four ReRAM dies via face-to-face bumps, achieving 14.08 to 135.69 token/s on a 55.98 mm<sup>2</sup> logic die.

### 28nm CNN-Transformer Accelerator for Semantic Segmentation

Nov. 2021 - Sept. 2024

AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China

- Contributed to the development of algorithm-hardware co-design optimizations for a 28nm CNN-Transformer semantic-segmentation accelerator, including hybrid attention processing, data-reuse-oriented layer fusion, and cascaded feature-map pruning for high-resolution ConvFormer workloads.
- Built a hardware energy simulation framework to quantify optimization impact and guide architecture and tape-out decisions; implemented the algorithm validation flow for VA/LA hybrid attention, KV/weight reuse scheduling, non-overlap layer fusion, and mask-based cascaded pruning.
- Designed RTL for attention/layer-fusion control and pruning-related datapath logic; assisted silicon testing and validation of the 13.93 mm<sup>2</sup> 28nm chip, achieving 0.22 uJ/token and up to 52.90 TOPS/W peak efficiency.

## PUBLICATIONS

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### **A 5nm 91.43 TOPS/W 4-Chiplet Generalizable-Rendering-Processor with UCIE-Enabled Cross-Die-Cache and Balance-Aware Progressive Multi-Level Sparsity**

Tan, Y.\* , Ma, S.\* , Dong, P., Luo, P., Lei, Z., Lu, W., Ying, G., ... & Cheng, K. T.

2026 IEEE Custom Integrated Circuits Conference (CICC), IEEE.

### **A 14.08-to-135.69Token/s ReRAM-on-Logic Stacked Outlier-Free Large-Language-Model Accelerator with Block-Clustered Weight-Compression and Adaptive Parallel-Speculative-Decoding**

Dong, P., Tan, Y., Liu, X., Luo, P., Liu, Y., Pang, D., Ma, S., ... & Cheng, K. T.

2026 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.

### **A 28nm 0.22uJ/Token Memory-Compute-Intensity-Aware CNN-Transformer Accelerator with Hybrid-Attention-Based Layer-Fusion and Cascaded Pruning for Semantic-Segmentation**

Dong, P.\* , Tan, Y.\* , Liu, X., Luo, P., Liu, Y., Liang, L., ... & Cheng, K. T.

2025 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.

### **APSQ: Additive Partial Sum Quantization with Algorithm-Hardware Co-Design**

Tan, Y.\* , Dong, P.\* , Wu, Y., Liu, Y., Liu, X., Luo, P., Liu, S. Y., Huang, X., Zhang, D., Liang, L., & Cheng, K. T.

2025 62nd ACM/IEEE Design Automation Conference (DAC), IEEE.

### **Genetic Quantization-Aware Approximation for Non-Linear Operations in Transformers**

Dong, P.\* , Tan, Y.\* , Zhang, D., Ni, T., Liu, X., Liu, Y., ... & Cheng, K. T.

2024 61st ACM/IEEE Design Automation Conference (DAC), IEEE.

### **A Reconfigurable Coprocessor for Simultaneous Localization and Mapping Algorithms in FPGA**

Tan, Y.\* , Deng, H.\* , Sun, M., Zhou, M., Chen, Y., Chen, L., ... & An, F.

IEEE Transactions on Circuits and Systems II: Express Briefs, 70(1), 286-290, 2022.

### **A Reconfigurable Visual-Inertial Odometry Accelerator with High Area and Energy Efficiency for Autonomous Mobile Robots**

Tan, Y.\* , Sun, M.\* , Deng, H., Wu, H., Zhou, M., Chen, Y., ... & An, F.

Sensors, 22(19), 7669, 2022.

\* Authors marked with an asterisk contributed equally to the corresponding work.

## HONORS AND AWARDS

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**Postgraduate Studentship (PGS) Award in HKUST**

Sept. 2023 - Present

**Best Teaching Assistant Award, Department of Electronic and Computer Engineering, HKUST**

Aug. 2025

**Outstanding Graduate (School Level), SUSTech**

May 2023

**First-Class Outstanding Students Scholarship with the highest score**

Sept. 2022

**Undergraduate Innovation and Entrepreneurship Training Program**

Apr. 2022

**Shenzhen Longsys Electronics Company Award (Top 2% in the School of Microelectronics)**

Dec. 2021

**First Prize, 2021 National College Students FPGA Innovation Design Competition (Top 22 out of 1,341 teams)**

Dec. 2021

**First Prize, 2021 International Competition of Autonomous Running Robots (1st place out of 34 finalist teams)**

Oct. 2021

## SKILLS

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Research Interests: Software/Hardware Co-Design, Model Compression, 3D Processing

Programming Languages: C, C++, Java, Python, SystemVerilog, Verilog HDL, VHDL

Professional Software: AutoCAD, Cadence, Design Compiler, IC Compiler II, MATLAB, Multisim, Silvaco

Languages: English (fluent), Mandarin (native), Cantonese (native)